

# TASK ON IAM

## 1. Create one IAM user and assign EC2 and S3 full access roles.

- Logged into the AWS Management Console and navigated to IAM service.
- Created a new IAM user named test-user.

The screenshot shows the 'Specify user details' step of the 'Create user' wizard in the AWS IAM console. On the left, a progress bar indicates three steps: Step 1 (Specify user details, selected), Step 2 (Set permissions), and Step 3 (Review and create). The main area is titled 'Specify user details' and contains a 'User details' section. The 'User name' field is filled with 'test-user'. Below the field, a note states: 'The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ \_ - (hyphen)'.

- Selected Attach policies directly option for assigning permissions.

The screenshot shows the 'Permissions options' section of the AWS IAM console. It contains three radio button options: 'Add user to group' (with a description: 'Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.'), 'Copy permissions' (with a description: 'Copy all group memberships, attached managed policies, and inline policies from an existing user.'), and 'Attach policies directly' (which is selected and highlighted with a blue border, with a description: 'Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.').

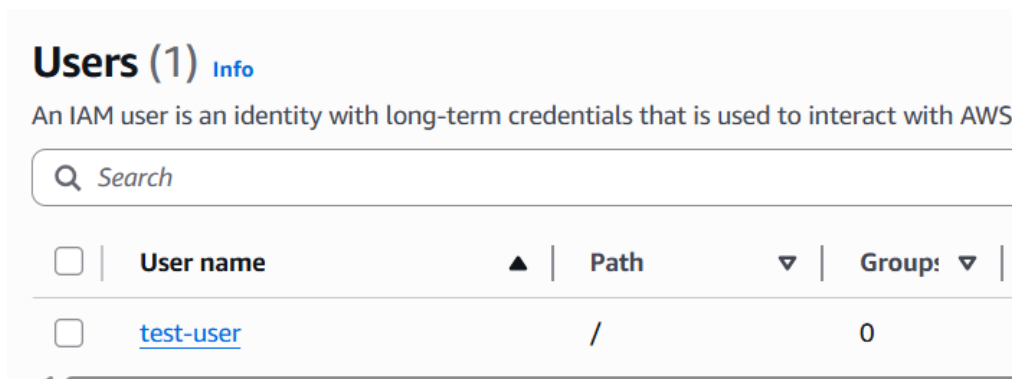
## Attached AWS managed policies:

- AmazonEC2FullAccess
- AmazonS3FullAccess

The screenshot shows the 'User details' and 'Permissions summary' sections of the AWS IAM console. The 'User details' section has three fields: 'User name' (test-user), 'Console password type' (None), and 'Require password reset' (No). The 'Permissions summary' section is a table with the following data:

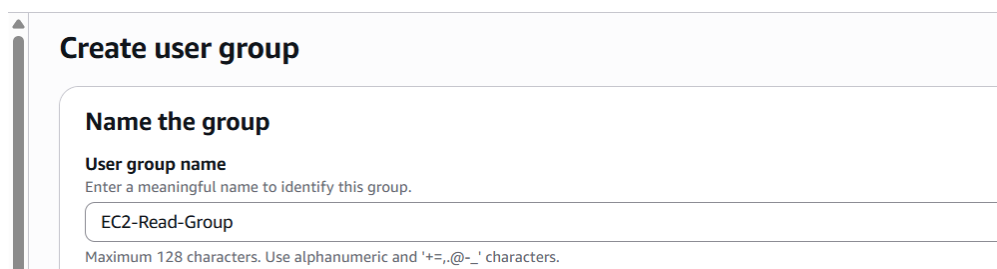
| Name                                | Type        | Used as            |
|-------------------------------------|-------------|--------------------|
| <a href="#">AmazonEC2FullAccess</a> | AWS managed | Permissions policy |
| <a href="#">AmazonS3FullAccess</a>  | AWS managed | Permissions policy |

- Verified permissions under Permissions Summary section.
- Successfully created the user with full access to EC2 and S3 services.

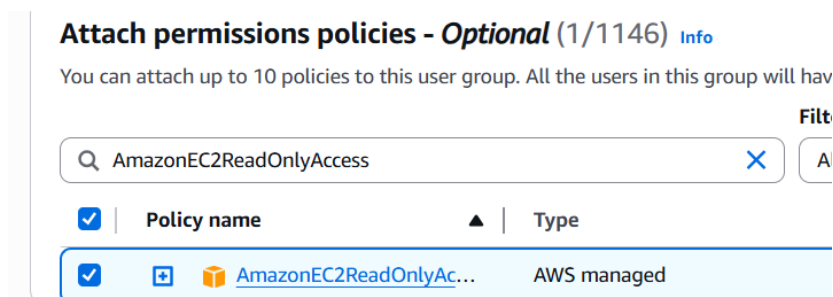


## 2. Create one group in IAM and assign read access for EC2.

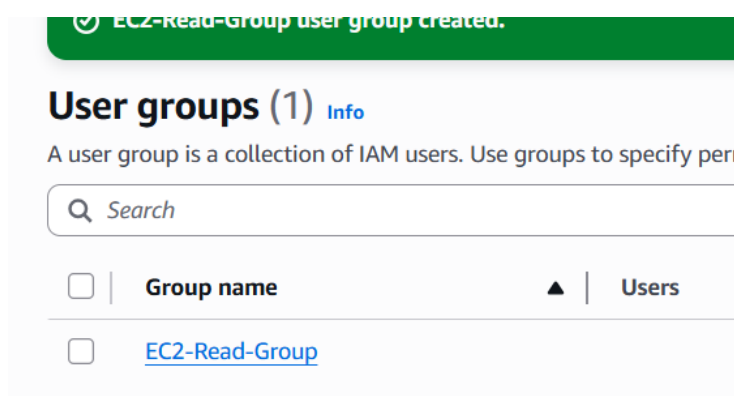
- Logged into the AWS Management Console and opened IAM service.
- Navigated to User Groups and selected Create group.



- Entered the group name as **EC2-Read-Group**
- In the permissions section, searched for **AmazonEC2ReadOnlyAccess** policy.



- Completed the group creation process successfully.



### 3. Create a new user named "Devops" and add to the group created in task 2.

- Logged into the AWS Management Console and navigated to IAM service.
- Selected Users → Create user option.
- Entered the user's name as Devops under user details section.

Step 1  
**Specify user details**

Step 2  
Set permissions

Step 3  
Review and create

### Specify user details

#### User details

**User name**

Devops

The user name can have up to 64 characters. Valid characters: A-Z

- Proceeded to Set permissions step and Add user to group option as EC2-Read-Group

### Set permissions

Add user to an existing group or create a new one. Using

#### Permissions options

☒ Add user to group  
Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

#### User groups (1/1)

Search

| <input checked="" type="checkbox"/> | Group name ↗   |
|-------------------------------------|----------------|
| <input checked="" type="checkbox"/> | EC2-Read-Group |

- New user Devops created successfully

### Users (2) [Info](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS.

Search

| <input type="checkbox"/> | User name                 | Path | Group: |
|--------------------------|---------------------------|------|--------|
| <input type="checkbox"/> | <a href="#">Devops</a>    | /    | 1      |
| <input type="checkbox"/> | <a href="#">test-user</a> | /    | 0      |

#### 4. Write a bash script to create an IAM user with VPC full access.

- Installed and configured AWS CLI using aws configure with valid credentials.

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ aws --version
aws-cli/2.34.30 Python/3.14.4 Windows/11 exe/AMD64
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ !
```

- Created a bash script file (create-user.sh) to automate IAM user creation.
- Executed the script using: ./create-user.sh

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ aws configure
AWS Access Key ID [*****VR0Z]: AKIA5QKV7AB3Q62GFDNU
AWS Secret Access Key [*****zhzj]: 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
Default region name [us-east-1]: us-east-1
Default output format [json]: json

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ nano create-user.sh

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ ./create-user.sh
{
  "User": {
    "Path": "/",
    "UserName": "vpc-user",
    "UserId": "AIDA5QKV7AB3U65W042WC",
    "Arn": "arn:aws:iam::928430096503:user/vpc-user",
    "CreateDate": "2026-04-20T11:01:41+00:00"
  }
}
User created successfully

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
```

```
GNU nano 6.3 create-user.sh Modified
#!/bin/bash

USERNAME="vpc-user"

# Create IAM user
aws iam create-user --user-name $USERNAME

# Attach VPC Full Access policy
aws iam attach-user-policy \
  --user-name $USERNAME \
  --policy-arn arn:aws:iam::aws:policy/AmazonVPCFullAccess

echo "User created successfully"
```

- Defined the IAM user name as **vpc-user** inside the script.
- Used AWS CLI command `aws iam create-user` to create the IAM user.

| <input type="checkbox"/> | User name                 | ▲ | Path | ▼ | Group! | ▼ | Last activity |
|--------------------------|---------------------------|---|------|---|--------|---|---------------|
| <input type="checkbox"/> | <a href="#">Devops</a>    |   | /    |   | 1      |   | -             |
| <input type="checkbox"/> | <a href="#">test-user</a> |   | /    |   | 0      |   | -             |
| <input type="checkbox"/> | <a href="#">vpc-user</a>  |   | /    |   | 0      |   | -             |

- **Attached AWS managed policy:**  
**AmazonVPCFullAccess using aws iam attach-user-policy**

**Permissions policies (1/1)** Remove

Permissions are defined by policies attached to the user directly or through groups.

Filter by Type

Q AmazonVPCFullAccess X All types 1 match

| <input checked="" type="checkbox"/> | Policy name ↗       | ▲ | Type        | ▼ | Attached via ↗ |
|-------------------------------------|---------------------|---|-------------|---|----------------|
| <input checked="" type="checkbox"/> | AmazonVPCFullAccess |   | AWS managed |   | Directly       |

- Confirmed that **AmazonVPCFullAccess** policy is attached to the user.

## 5. Create an IAM policy to allow EC2 access for a specific user in specific regions only.

- Logged into the AWS Management Console and opened IAM service.
- Navigated to Policies → Create Policy.
- Selected JSON tab to define custom policy.

IAM > Policies > Create policy

Step 1  
● **Specify permissions**

Step 2  
○ Review and create

**Specify permissions** Info

Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.

**Policy editor** Visual **JSON**

```

1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Effect": "Allow",
6       "Action": "ec2:*",
7       "Resource": "*",
8       "Condition": {
9         "StringEquals": {
10          "aws:RequestedRegion": "us-east-1"
11        }
12      }
13    }
14  ]
15 }
```

Edit statement

**Add actions**

Choose a service

Q Search serv

- Created a policy allowing EC2 actions using:
- "Action": "ec2:\*" and "Resource": "\*"
- Added a Condition block to restrict access to a specific region.
- Used condition key: "aws:RequestedRegion": "us-east-1"
- Named the policy as EC2-Region-Restricted.
- Created the policy successfully.

## Review and create [Info](#)

Review the permissions, specify details, and tags.

### Policy details

**Policy name**  
Enter a meaningful name to identify this policy.

Maximum 128 characters. Use alphanumeric and '+,=, @, -, \_' characters.

- Attached the policy to the IAM user.

|                                     |                                                         |   |                  |   |                   |
|-------------------------------------|---------------------------------------------------------|---|------------------|---|-------------------|
| Q                                   | EC2-Region-Restricted                                   | X | All types        | ▼ | 1 match           |
| <input checked="" type="checkbox"/> | Policy name <a href="#">↗</a>                           | ▲ | Type             | ▼ | Attached entities |
| <input checked="" type="checkbox"/> | <a href="#">+</a> <a href="#">EC2-Region-Restricted</a> |   | Customer managed |   | 0                 |

- Verified that EC2 access is allowed only in the test-user.

|                          |                                                                                                                                               |                |                  |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------|------------------|
| Q                        | Search                                                                                                                                        | Filter by Type | All types        |
| <input type="checkbox"/> | Policy name <a href="#">↗</a>                                                                                                                 | ▲              | Type             |
| <input type="checkbox"/> | <a href="#">+</a>  <a href="#">AmazonEC2FullAccess</a>     |                | AWS managed      |
| <input type="checkbox"/> | <a href="#">+</a>  <a href="#">AmazonEC2ReadOnlyAccess</a> |                | AWS managed      |
| <input type="checkbox"/> | <a href="#">+</a>  <a href="#">AmazonS3FullAccess</a>      |                | AWS managed      |
| <input type="checkbox"/> | <a href="#">+</a> <a href="#">EC2-Region-Restricted</a>                                                                                       |                | Customer managed |

6. We have two accounts: Account A and Account B. Account A user should access an S3 bucket in Account B.

(Collaborate with a team member and execute this. This is mostly asked in every interview.)

- Collaborated with a team member using two AWS accounts.

≡ [Amazon S3](#) > [Buckets](#) > cross-account-demo-bucket145

 Successfully created bucket "cross-account-demo-bucket145"

### cross-account-demo-bucket145 [Info](#)

[Objects](#) | [Metadata](#) | [Properties](#) | [Permissions](#) | [Metrics](#) | [Management](#) | [File systems - new](#)

- Created an S3 bucket in Account B.

- Configured bucket policy to allow access to IAM user from Account A.

**Devops** [Info](#)

**Summary**

|                                                                                                                                |                            |                                             |
|--------------------------------------------------------------------------------------------------------------------------------|----------------------------|---------------------------------------------|
| ARN<br> arn:aws:iam::928430096503:user/Devops | Console access<br>Disabled | Access key<br><a href="#">Create acc...</a> |
| Created<br>April 20, 2026, 15:47 (UTC+05:30)                                                                                   | Last console sign-in<br>-  |                                             |

- Defined Principal using IAM user ARN.

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowListBucket",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::009365819224:user/test-user"
      },
      "Action": "s3:ListBucket",
      "Resource": "arn:aws:s3:::cross-account-demo-bucket145"
    },
    {
      "Sid": "AllowObjectAccess",
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::009365819224:user/test-user"
      },
      "Action": [
        "s3:GetObject",

```

- Granted required S3 permissions (ListBucket, GetObject)
- Attached IAM permissions to user in Account A.

**Policy editor**

```

1  {
2    "Version": "2012-10-17",
3    "Statement": [
4      {
5        "Sid": "ListBucket",
6        "Effect": "Allow",
7        "Action": "s3:ListBucket",
8        "Resource": "arn:aws:s3:::cross-account-demo-bucket0408"
9      },
10     {
11       "Sid": "GetObject",
12       "Effect": "Allow",
13       "Action": "s3:GetObject",
14       "Resource": "arn:aws:s3:::cross-account-demo-bucket0408/*"
15     }
16   ]
17 }

```

- Verified access using AWS CLI (aws s3 ls).

```
Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ aws configure
AWS Access Key ID [*****FDNU]: AKIA5QKV7AB3XV2AZJWD
AWS Secret Access Key [*****sXw]: 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
Default region name [us-east-1]: us-east-1
Default output format [json]: json

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ aws sts get-caller-identity
{
  "UserId": "AIDA5QKV7AB3TBVIT2KUT",
  "Account": "928430096503",
  "Arn": "arn:aws:iam::928430096503:user/Devops"
}

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$ aws s3 ls s3://cross-account-demo-bucket0408
2026-04-20 18:36:54          293 create_user.bash

Arunsai@DESKTOP-N6MPCBS MINGW64 ~/Downloads
$
```

- Successfully accessed cross-account S3 resources.

### Permissions policies (2)

Permissions are defined by policies attached to the user directly or through groups.

Filter by Typ All types

| <input type="checkbox"/> | Policy name <a href="#">↗</a>                                                                                                                                                                                   | ▲ | Type            |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-----------------|
| <input type="checkbox"/> |   <a href="#">AmazonEC2ReadOnlyAccess</a> |   | AWS managed     |
| <input type="checkbox"/> |  <a href="#">Collobrate</a>                                                                                                  |   | Customer inline |